

## LIST OF EXPERIMENTS

1. **Milgram Experiment:** A study on obedience to authority figures, conducted by Stanley Milgram in the 1960s.
2. **Stanford Prison Experiment:** A study of the psychological effects of becoming a prisoner or prison guard, conducted by Philip Zimbardo in 1971.
3. **Pavlov's Dogs:** A study on classical conditioning, conducted by Ivan Pavlov in the 1890s.
4. **Asch Conformity Experiment:** A study on conformity, conducted by Solomon Asch in the 1950s.
5. **Hawthorne Effect:** A study on the effect of changes in working conditions on productivity, conducted by Elton Mayo in the 1920s.
6. **Bobo Doll Experiment:** A study on the effects of observed aggression, conducted by Albert Bandura in the 1960s.
7. **Little Albert Experiment:** A study on classical conditioning of fear, conducted by John B. Watson and Rosalie Rayner in 1920.
8. **The Marshmallow Test:** A study on delayed gratification, conducted by Walter Mischel in the 1960s.
9. **The Invisible Gorilla:** A study on inattention blindness, conducted by Christopher Chabris and Daniel Simons in 1999.
10. **The Bystander Effect:** A study on the diffusion of responsibility, conducted by John Darley and Bibb Latané in the 1960s.

**Note: A minimum of ten experiments from the following should be performed.**

1. Experiment 1: Description of experiment 1.
2. Experiment 2: Description of experiment 2.
3. Experiment 3: Description of experiment 3.
4. Experiment 4: Description of experiment 4.
5. Experiment 5: Description of experiment 5.
6. Experiment 6: Description of experiment 6.
7. Experiment 7: Description of experiment 7.
8. Experiment 8: Description of experiment 8.
9. Experiment 9: Description of experiment 9.
10. Experiment 10: Description of experiment 10.

### (A) Hardware based experiments

Hardware-based experiments are experiments that involve the use of physical equipment or devices to conduct research. These experiments can be used to test the performance, reliability, and functionality of hardware components, systems, and devices.

Some examples of hardware-based experiments include:

1. Testing the performance of computer processors, memory, and storage devices.
2. Evaluating the reliability and durability of electronic components and systems.
3. Measuring the accuracy and precision of sensors and measurement devices.
4. Assessing the functionality and usability of consumer electronics and appliances.

Hardware-based experiments can provide valuable insights into the design and development of new technologies, as well as help to improve the performance and reliability of existing systems. These experiments can be conducted in a controlled laboratory environment, or in real-world settings to gather data and information under realistic conditions.

### 1. Verification of Kirchhoff's laws

Kirchhoff's laws are two laws that deal with the conservation of charge and energy in electrical circuits. These laws are known as Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL).

- **Kirchhoff's Current Law (KCL):** This law states that the algebraic sum of currents entering and leaving a node in an electrical circuit is zero. In other words, the total current entering a node is equal to the total current leaving the node.
- **Kirchhoff's Voltage Law (KVL):** This law states that the algebraic sum of the voltage drops around any closed loop in an electrical circuit is zero. In other words, the total voltage rise in a closed loop is equal to the total voltage drop in the same loop.

To verify Kirchhoff's laws, an experiment can be performed using a simple electrical circuit consisting of a battery, resistors, and a multimeter. The multimeter can be used to measure the current and voltage at different points in the circuit. By comparing the measured values with the values predicted by Kirchhoff's laws, the validity of the laws can be verified.

### 2. Measurement of power and power factor in a single phase ac series inductive circuit and study improvement of power factor using capacitor

- Power is the rate at which energy is transferred or used. In an AC circuit, power can be calculated by multiplying the voltage and current, and then multiplying by the power factor.
- The power factor is the ratio of the real power to the apparent power. It is a measure of how effectively the current is being used to produce real power.
- In a single phase AC series inductive circuit, the power factor is typically less than 1 due to the presence of inductive reactance, which causes the

current to lag behind the voltage.

- The power factor can be improved by adding a capacitor in parallel with the inductive load. The capacitor provides capacitive reactance, which causes the current to lead the voltage, thus reducing the phase difference between the voltage and current.
- By reducing the phase difference, the power factor is improved, resulting in more efficient use of the current and a reduction in power losses.
- The power factor can be measured using a power factor meter, which measures the phase difference between the voltage and current and calculates the power factor.
- The power factor can also be calculated by measuring the real power and apparent power and dividing the real power by the apparent power.
- Improving the power factor in a single phase AC series inductive circuit can result in more efficient use of power and reduced power losses. This can lead to cost savings and improved system performance.

### **3. Study of phenomenon of resonance in RLC series circuit and obtain resonant frequency.**

- Resonance is a phenomenon that occurs in RLC series circuits when the inductive reactance and capacitive reactance are equal in magnitude.
- At resonance, the impedance of the circuit is purely resistive, and the current in the circuit is at its maximum value.
- The resonant frequency of an RLC series circuit is given by the formula:  $f = 1 / (2 * \pi * \sqrt{L * C})$ , where  $f$  is the resonant frequency,  $L$  is the inductance, and  $C$  is the capacitance.
- To obtain the resonant frequency of an RLC series circuit, the values of the inductance and capacitance must be known.
- The resonant frequency can be determined experimentally by measuring the current in the circuit as the frequency of the applied voltage is varied.
- At the resonant frequency, the current in the circuit will be at its maximum value.
- The phenomenon of resonance has many practical applications, such as in the tuning of radio receivers and the design of filters in electronic circuits.

### **4. Connection and measurement of power consumption of a fluorescent lamp (tube light)**

A fluorescent lamp, also known as a tube light, is a low-pressure mercury-vapor gas-discharge lamp that uses fluorescence to produce visible light. To measure the power consumption of a fluorescent lamp, the following steps can be taken:

1. **Connect the lamp to a power source:** The fluorescent lamp should be connected to a power source using the appropriate wiring and connections. Ensure that the connections are secure and that the lamp is properly installed.

2. **Measure the current:** Use a clamp meter to measure the current flowing through the circuit. The clamp meter should be placed around one of the wires leading to the lamp.
3. **Measure the voltage:** Use a multimeter to measure the voltage across the lamp. The multimeter should be connected to the two wires leading to the lamp.
4. **Calculate the power consumption:** The power consumption of the lamp can be calculated by multiplying the current and voltage measurements. The result will be the power consumption in watts.

It is important to note that the power consumption of a fluorescent lamp may vary depending on factors such as the age of the lamp, the temperature of the environment, and the quality of the power supply. Regularly measuring the power consumption of a fluorescent lamp can help to identify any changes in its performance and efficiency.

#### **5. Measurement of power in 3- phase circuit by two-wattmeter method and determination of its power factor for star as well as delta connected load.**

The two-wattmeter method is used to measure the power in a three-phase circuit. This method is applicable for both balanced and unbalanced loads, and for both star and delta connected loads.

The procedure for measuring power using the two-wattmeter method is as follows: 1. Connect the two wattmeters as shown in the diagram, with one wattmeter connected between phase A and the neutral, and the other wattmeter connected between phase B and the neutral. 2. The readings of the two wattmeters are then added to obtain the total power consumed by the load. 3. The power factor of the load can be determined by using the formula: power factor =  $(W1 - W2) / (W1 + W2)$ , where W1 and W2 are the readings of the two wattmeters.

This method is applicable for both star and delta connected loads. For a star connected load, the neutral point of the load is connected to the neutral of the supply, while for a delta connected load, the neutral point of the load is not connected to the neutral of the supply.

In summary, the two-wattmeter method is a simple and effective way to measure the power in a three-phase circuit, and to determine the power factor of the load. It is applicable for both balanced and unbalanced loads, and for both star and delta connected loads.

#### **6. Determination of parameters of ac single phase series RLC circuit**

An RLC circuit is an electrical circuit consisting of a resistor (R), an inductor (L), and a capacitor (C), connected in series or in parallel. The name of the circuit is derived from the letters that are used to denote the constituent components of this circuit, where the sequence of the components may vary from RLC.

The circuit forms a harmonic oscillator for current and resonates similarly to an LC circuit. The main difference that the presence of the resistor makes is that any oscillation induced in the circuit will die away over time if it is not kept going by a source. This effect of the resistor is called damping. The presence of the resistor also reduces the peak resonant frequency somewhat.

The behavior of the circuit is governed by the differential equation:

$$L * (d^2 * q(t)) / dt^2 + R * (dq(t)) / dt + (1/C) * q(t) = V(t)$$

where  $q(t)$  is the electric charge on the capacitor as a function of time  $t$ , and  $V(t)$  is the voltage source driving the circuit.

The parameters of an AC single-phase series RLC circuit can be determined by measuring the voltage and current at different points in the circuit and using Ohm's law and Kirchhoff's circuit laws to calculate the values of the resistance, inductance, and capacitance.

Some of the methods used to determine the parameters of an AC single-phase series RLC circuit include: 1. Impedance measurement: The impedance of the circuit can be measured by applying an AC voltage to the circuit and measuring the resulting current. The impedance can then be calculated using Ohm's law, and the values of the resistance, inductance, and capacitance can be determined from the impedance. 2. Resonant frequency measurement: The resonant frequency of the circuit can be determined by measuring the frequency at which the voltage across the capacitor or the current through the inductor is at a maximum. The values of the inductance and capacitance can then be calculated using the formula for the resonant frequency of an RLC circuit. 3. Time constant measurement: The time constant of the circuit can be determined by measuring the time it takes for the voltage across the capacitor to decay to a certain fraction of its initial value after the circuit is disconnected from the voltage source. The values of the resistance and capacitance can then be calculated using the formula for the time constant of an RC circuit.

These are some of the methods that can be used to determine the parameters of an AC single-phase series RLC circuit. It is important to note that the accuracy of the results obtained using these methods depends on the accuracy of the measurements and the assumptions made in the calculations. It is always a good idea to use multiple methods and compare the results to ensure the accuracy of the determined parameters.

## **7. Determination of (i) Voltage ratio (ii) polarity and (iii) efficiency by load test of a single phase Transformer**

A single-phase transformer is an electrical device that is used to transfer electrical energy from one circuit to another by electromagnetic induction. The voltage ratio, polarity, and efficiency of a single-phase transformer can be determined by performing a load test.

1. **Voltage Ratio:** The voltage ratio of a transformer is the ratio of the secondary voltage to the primary voltage. It can be determined by measuring the primary and secondary voltages of the transformer while it is under load and calculating the ratio.
2. **Polarity:** The polarity of a transformer refers to the direction of the induced voltage in the secondary winding relative to the primary winding. It can be determined by performing a polarity test, which involves connecting the primary and secondary windings in series and measuring the resulting voltage.
3. **Efficiency:** The efficiency of a transformer is the ratio of the output power to the input power. It can be determined by measuring the input and output power of the transformer while it is under load and calculating the ratio.

During a load test, the transformer is connected to a load and the primary and secondary voltages, as well as the input and output power, are measured. These measurements can then be used to calculate the voltage ratio, polarity, and efficiency of the transformer. It is important to note that the load test should be performed under normal operating conditions to obtain accurate results.

## 8. Determination of efficiency of a dc shunt motor by load test

The efficiency of a DC shunt motor can be determined by conducting a load test. The load test involves applying a load to the motor and measuring its performance under the load. The following steps can be followed to determine the efficiency of a DC shunt motor by load test:

1. Connect the DC shunt motor to a suitable load, such as a brake or a dynamometer.
2. Measure the input power to the motor by measuring the voltage and current supplied to the motor.
3. Measure the output power of the motor by measuring the torque and speed of the motor.
4. Calculate the efficiency of the motor using the formula:  $\text{Efficiency} = (\text{Output Power} / \text{Input Power}) * 100\%$
5. Repeat the above steps for different loads and plot the efficiency vs. load curve.

The efficiency of the motor can be determined from the efficiency vs. load curve. The maximum efficiency of the motor occurs at the point where the efficiency vs. load curve is maximum. The efficiency of the motor decreases as the load on the motor increases.

**9. To study running and speed reversal of a three phase induction motor and record speed in both directions.**

A three-phase induction motor is an AC motor that operates on the principle of electromagnetic induction. The motor consists of a stator, which is a stationary part, and a rotor, which is a rotating part. The stator has a three-phase winding, and when a three-phase AC supply is given to it, a rotating magnetic field is produced. This rotating magnetic field induces an electromotive force in the rotor, which causes the rotor to rotate.

The speed of a three-phase induction motor can be controlled by varying the frequency of the AC supply given to the stator. The speed of the motor is directly proportional to the frequency of the AC supply. Therefore, by increasing the frequency of the AC supply, the speed of the motor can be increased, and by decreasing the frequency of the AC supply, the speed of the motor can be decreased.

The direction of rotation of a three-phase induction motor can be reversed by interchanging any two of the three-phase supply connections to the stator. This changes the direction of the rotating magnetic field, which in turn changes the direction of rotation of the rotor.

To study the running and speed reversal of a three-phase induction motor, the following steps can be followed:

1. Connect the three-phase induction motor to a variable frequency drive (VFD).
2. Set the frequency of the VFD to a desired value and observe the speed of the motor.
3. Increase or decrease the frequency of the VFD to change the speed of the motor and record the speed at different frequencies.
4. Interchange any two of the three-phase supply connections to the stator to reverse the direction of rotation of the motor.
5. Observe and record the speed of the motor in the reverse direction at different frequencies.

By following these steps, the running and speed reversal of a three-phase induction motor can be studied and the speed of the motor can be recorded in both directions.

**10. Demonstration of cut-out sections of machines: dc machine, three phase induction machine, single-phase induction machine and synchronous machine.**

A cut-out section of a machine is a physical model that shows the internal components and workings of the machine. It is created by cutting away a portion of the machine's casing to reveal the internal components. Cut-out sections are commonly used for educational and demonstration purposes.

1. **DC Machine:** A DC machine is an electrical machine that converts mechanical energy into direct current (DC) electrical energy or vice versa. A cut-out section of a DC machine would typically show the armature, commutator, brushes, and field windings.
2. **Three Phase Induction Machine:** A three phase induction machine is an AC electric motor that converts electrical energy into mechanical energy. A cut-out section of a three phase induction machine would typically show the stator, rotor, and windings.
3. **Single-Phase Induction Machine:** A single-phase induction machine is an AC electric motor that operates on single-phase power. A cut-out section of a single-phase induction machine would typically show the stator, rotor, and windings.
4. **Synchronous Machine:** A synchronous machine is an AC electric machine that operates at a constant speed, determined by the frequency of the power source. A cut-out section of a synchronous machine would typically show the rotor, stator, and field windings.

These cut-out sections provide a visual representation of the internal workings of the machines, allowing for a better understanding of their operation and construction. They are commonly used in educational settings to demonstrate the principles of electromechanical energy conversion.

## (B) Experiments available on virtual lab

Virtual labs provide a platform for students to perform experiments and learn in a simulated environment. There are many virtual labs available that offer a wide range of experiments for students to perform. Here are some examples of virtual labs and the experiments they offer:

1. **Virtual Chemistry Experiments** - A collection of interactive web-based chemistry tutorials .
2. **PraxiLabs** - Offers virtual experiments that adapt to any learning style, including progress tracking, walkthrough videos, lab manuals, enrichment documents, immediate feedback, MCQ quizzes, and much more .
3. **VRLab Academy** - Offers more than 200 science experiments available on VR and PC .
4. **American Chemical Society** - Offers virtual chemistry and simulations, including the chemistry of aspirin, average atomic mass, solubility with rock candy, and freezing point depression with road salt .
5. **Mystery Science** - Offers a wealth of virtual science labs, activities, and experiments for students in elementary school, covering topics such as weather, astronomy, humans, and animals .
6. **National Science Digital Library** - Offers ChemCollective Virtual Labs, which provide a deep dive into chemistry .



These are just a few examples of the many virtual labs available that offer a wide range of experiments for students to perform and learn from.

### **1. Kirchhoff's laws**

Kirchhoff's laws are two laws that deal with the conservation of charge and energy in electrical circuits. These laws are named after the German physicist Gustav Kirchhoff.

1. Kirchhoff's Current Law (KCL): This law states that the algebraic sum of currents entering and leaving a node in an electrical circuit is zero. In other words, the total current entering a node is equal to the total current leaving the node. This law is based on the principle of conservation of electric charge.
2. Kirchhoff's Voltage Law (KVL): This law states that the algebraic sum of the voltage drops around any closed loop in an electrical circuit is zero. In other words, the total voltage rise in a closed loop is equal to the total voltage drop in the same loop. This law is based on the principle of conservation of energy.

These laws are widely used in circuit analysis to determine the current and voltage in different parts of a circuit. They are fundamental principles that must be satisfied in any electrical circuit.

### **Virtual Lab**

Virtual Lab at Amrita uses state-of-the-art computer simulation technology to create real-world environments and problem handling capabilities. It is required to bridge the gap between institutions (or industries) that retain the physical laboratory and distantly placed economically challenged educational institutions in areas such as Physics, Chemistry, and Biotechnology .

Virtual labs are media-rich online learning environments, which allow users to conduct physical laboratory experiments in a computer-simulated environment. This innovative new educational technique is enabling experimentation beyond the boundaries of traditional laboratories .

The link you provided is for the Kirchhoff's Laws simulation in the Amrita Vishwa Vidyapeetham Virtual Lab. This simulation allows users to learn about Kirchhoff's Voltage Law, which states that "The algebraic sum of all voltages in a loop must equal zero" .

### **2. Thevenin Theorem**

Thevenin's Theorem is a technique used to analyze complex electrical circuits by simplifying them into an equivalent circuit composed of a single voltage source, a single resistor, and a load. The theorem states that any linear, bilateral circuit

can be replaced by an equivalent circuit consisting of a voltage source and a series resistor.

The steps to apply Thevenin's Theorem are as follows: 1. Remove the load resistor from the original circuit. 2. Calculate the voltage across the open connection where the load resistor was removed. This voltage is known as the Thevenin voltage ( $V_{th}$ ). 3. Calculate the equivalent resistance of the circuit as seen from the open connection where the load resistor was removed. This resistance is known as the Thevenin resistance ( $R_{th}$ ). 4. Replace the original circuit with an equivalent circuit consisting of a voltage source with a voltage equal to the Thevenin voltage ( $V_{th}$ ) in series with a resistor with a resistance equal to the Thevenin resistance ( $R_{th}$ ). 5. Reconnect the load resistor to the equivalent circuit.

Thevenin's Theorem can be used to analyze both DC and AC circuits. It is a powerful tool for simplifying complex circuits and making them easier to analyze. It is also useful for analyzing circuits with multiple sources and for determining the maximum power transfer to a load.

### Virtual Lab

Virtual labs are online tools that offer a space for students to engage with their subject matter interactively. They can be used for pre-lab preparation, online lab courses, post-lab review, make-up labs, or used with a textbook to bridge the gap between lab and lecture courses. Virtual labs are available in various disciplines of Science and Engineering, including Anatomy & Physiology, Biology (Majors and Non-majors), Botany, Environmental Science, Genetics, Human Biology, Medical Assisting, Microbiology, and Nutrition.

Virtual labs are designed to enthruse students to conduct experiments by arousing their curiosity. This helps them in learning basic and advanced concepts through remote experimentation. Some virtual labs are even available for free and include built-in quizzes, notebooks, and references.

Virtual labs are a great way to provide remote-access to simulation-based Labs and improve the quality of care and education programs. They are a valuable resource for faculty and students in the remote learning context.

### 3. RLC series resonance

- An RLC series circuit consists of a resistor ( $R$ ), an inductor ( $L$ ), and a capacitor ( $C$ ) connected in series.
- At resonance, the impedance of the circuit is at its minimum value, and the current flowing through the circuit is at its maximum value.
- The resonant frequency of the circuit is given by the formula:  $f = 1 / (2 * \pi * \sqrt{L * C})$ .
- The quality factor ( $Q$ ) of the circuit is given by the formula:  $Q = (1 / R) * \sqrt{L / C}$ .

- The bandwidth of the circuit is given by the formula:  $BW = f / Q$ .
- The voltage across the resistor, inductor, and capacitor are in phase at resonance.
- The voltage across the inductor leads the current by 90 degrees, and the voltage across the capacitor lags the current by 90 degrees.
- The total impedance of the circuit is given by the formula:  $Z = R + j * (X_L - X_C)$ , where  $X_L = 2 * \pi * f * L$  is the inductive reactance, and  $X_C = 1 / (2 * \pi * f * C)$  is the capacitive reactance.
- The phase angle of the circuit is given by the formula:  $\phi = \text{atan}((X_L - X_C) / R)$ .
- The power factor of the circuit is given by the formula:  $PF = \cos(\phi)$ .

### Virtual Lab

Virtual labs are online tools that offer a space for students to engage with their subject matter interactively. They can be used for pre-lab preparation, online lab courses, post-lab review, make-up labs, or used with a textbook to bridge the gap between lab and lecture courses.

Some benefits of virtual labs include: - They provide remote access to simulation-based labs in various disciplines of Science and Engineering. - They can help enthuse students to conduct experiments by arousing their curiosity, helping them learn basic and advanced concepts through remote experimentation. - They can be used in a remote learning context, making them accessible to students who may not have access to physical labs.

Virtual labs are available in a variety of disciplines, including Anatomy & Physiology, Biology (Majors and Non-majors), Botany, Environmental Science, Genetics, Human Biology, Medical Assisting, Microbiology, and Nutrition. They often include built-in quizzes, notebooks, and references to help students learn and retain information.

### 4. Measurement of power in 3- phase circuit by two wattmeter method and determination of its power factor for star as well as delta connected load.

The two wattmeter method is a common method used to measure power in a three-phase circuit. This method involves the use of two wattmeters to measure the power consumed by a three-phase load. The power factor of the load can also be determined using this method.

The two wattmeter method can be used for both star and delta connected loads. The procedure for measuring power and determining the power factor is slightly different for each type of connection.

For a star connected load, the two wattmeters are connected between the line and the neutral. The readings of the two wattmeters are then added together to obtain the total power consumed by the load. The power factor can be

determined by dividing the total power by the product of the line voltage and the line current.

For a delta connected load, the two wattmeters are connected between the lines. The readings of the two wattmeters are then added together to obtain the total power consumed by the load. The power factor can be determined by dividing the total power by the product of the line voltage and the line current.

In summary, the two wattmeter method is a useful and accurate method for measuring power in a three-phase circuit and determining its power factor for both star and delta connected loads. It is important to follow the correct procedure for the type of load being measured to ensure accurate results.

### Virtual Lab

Virtual Lab is an online platform that provides students with the opportunity to conduct experiments and learn about various scientific concepts in a virtual environment. The link provided leads to a virtual lab on the topic of measurement, hosted by the Virtual Labs project at the Dayalbagh Educational Institute.

In this virtual lab, students can learn about the following topics:

1. The importance of measurement in scientific experiments.
2. The different units of measurement and their conversions.
3. The use of various measuring instruments such as rulers, vernier calipers, and micrometers.
4. The concept of precision and accuracy in measurement.
5. The proper techniques for taking measurements and recording data.

This virtual lab provides an interactive and engaging learning experience for students, allowing them to gain a deeper understanding of the topic of measurement. It is a valuable resource for students preparing for exams or for anyone interested in learning more about this fundamental scientific concept.

### 5. Determination of parameters of ac single phase series RLC circuit

An AC single-phase series RLC circuit consists of a resistor (R), an inductor (L), and a capacitor (C) connected in series with an AC voltage source. The parameters of this circuit can be determined using the following steps:

1. **Impedance:** The total impedance (Z) of the circuit can be calculated using the formula  $Z = R + j(X_L - X_C)$ , where  $X_L$  is the inductive reactance and  $X_C$  is the capacitive reactance.  $X_L$  can be calculated as  $X_L = 2\pi fL$ , where  $f$  is the frequency of the AC source and  $L$  is the inductance.  $X_C$  can be calculated as  $X_C = 1/(2\pi fC)$ , where  $C$  is the capacitance.
2. **Current:** The current (I) flowing through the circuit can be calculated using Ohm's law,  $I = V/Z$ , where  $V$  is the voltage of the AC source.

3. **Voltage:** The voltage across each component can be calculated using Ohm's law. The voltage across the resistor (VR) is  $VR = IR$ , the voltage across the inductor (VL) is  $VL = IXL$ , and the voltage across the capacitor (VC) is  $VC = IXC$ .
4. **Power:** The average power (P) consumed by the circuit can be calculated using the formula  $P = I^2R$ .
5. **Phase angle:** The phase angle ( ) between the voltage and the current can be calculated using the formula  $\tan( ) = (XL - XC)/R$ .

These are the basic steps to determine the parameters of an AC single-phase series RLC circuit. It is important to note that the values of the parameters may vary depending on the frequency of the AC source.

### Virtual Lab

Virtual labs are online tools that offer a space for students to engage with their subject matter interactively. They can be used for pre-lab preparation, online lab courses, post-lab review, make-up labs, or used with a textbook to bridge the gap between lab and lecture courses . Virtual labs are available in various disciplines of Science and Engineering, including Anatomy & Physiology, Biology (Majors and Non-majors), Botany, Environmental Science, Genetics, Human Biology, Medical Assisting, Microbiology, and Nutrition . They are designed to enthruse students to conduct experiments by arousing their curiosity and help them in learning basic and advanced concepts through remote experimentation .

Some benefits of virtual labs include: - They provide remote-access to simulation-based Labs. - They can be used in the remote learning context. - They offer a robust online professional development system. - They include built-in quizzes, notebooks, and references.

### 6. To observe the B-H loop of a ferromagnetic material in CRO.

- The B-H loop is a graphical representation of the relationship between the magnetic field strength (H) and the magnetic flux density (B) of a ferromagnetic material.
- The B-H loop is also known as the hysteresis loop.
- A ferromagnetic material is a material that exhibits strong magnetism in the presence of an external magnetic field.
- A CRO (Cathode Ray Oscilloscope) is an electronic instrument that can be used to display and analyze the waveform of electronic signals.
- To observe the B-H loop of a ferromagnetic material in a CRO, the material is subjected to an alternating magnetic field.
- The CRO displays the B-H loop as a closed curve on the screen.
- The shape of the B-H loop provides information about the magnetic properties of the ferromagnetic material, such as its coercivity and remanence.

- Coercivity is the measure of the magnetic field strength required to reduce the magnetization of a ferromagnetic material to zero.
- Remanence is the measure of the residual magnetization of a ferromagnetic material after the external magnetic field has been removed.
- Observing the B-H loop of a ferromagnetic material in a CRO is an important technique in the study of magnetism and magnetic materials.

### **Virtual Lab**

Virtual labs are online tools that offer a space for students to engage with their subject matter interactively. They can be used for pre-lab preparation, online lab courses, post-lab review, make-up labs, or used with a textbook to bridge the gap between lab and lecture courses. Virtual labs are available in various disciplines of Science and Engineering, including Anatomy & Physiology, Biology (Majors and Non-majors), Botany, Environmental Science, Genetics, Human Biology, Medical Assisting, Microbiology, and Nutrition. They are designed to enthrall students to conduct experiments by arousing their curiosity and help them in learning basic and advanced concepts through remote experimentation.

Some of the benefits of virtual labs include: - They provide remote-access to simulation-based Labs in various disciplines of Science and Engineering. - They offer a robust online professional development system. - They include built-in quizzes, notebooks, and references.

Virtual labs are available as stand-alone products or can be assignable within certain educational platforms such as McGraw Hill ALEKS and Connect. They are available for high school and college students.

### **7. Determination of the efficiency of a dc motor by loss summation method (Swinburne's test)**

Swinburne's test is a method used to determine the efficiency of a DC motor. This test is performed by measuring the losses in the motor and then calculating the efficiency using the loss summation method. The following are the steps involved in performing Swinburne's test:

1. The first step is to measure the no-load losses of the motor. This is done by running the motor at its rated speed without any load connected to it. The power input to the motor is measured using a wattmeter.
2. The next step is to measure the copper losses in the armature and field windings. This is done by measuring the current flowing through the windings and the resistance of the windings.
3. Once the no-load losses and copper losses have been measured, the efficiency of the motor can be calculated using the loss summation method. This method involves adding up all the losses in the motor and then subtracting them from the input power to determine the output power.

4. The efficiency of the motor can then be calculated by dividing the output power by the input power and multiplying by 100 to express the result as a percentage.

Swinburne's test is a simple and effective method for determining the efficiency of a DC motor. It is commonly used in laboratory settings to evaluate the performance of motors. You can learn more about this test by visiting the virtual lab link provided.

### **Course Outcomes:**

- A course outcome is a statement that describes the knowledge, skills, and abilities that students are expected to demonstrate upon completion of a course.
- Course outcomes are used to guide the development of course content, assessments, and teaching methods.
- They provide a clear and measurable way to evaluate student learning and the effectiveness of the course.
- Course outcomes should be specific, measurable, achievable, relevant, and time-bound.
- They should be aligned with program and institutional goals, as well as with professional standards and expectations.
- Course outcomes should be communicated to students at the beginning of the course and should be used to guide their learning throughout the course.
- The assessment of course outcomes should be an ongoing process, with regular feedback provided to students to help them improve their performance.
- The results of course outcome assessments can be used to inform course and program improvements, as well as to demonstrate the effectiveness of the course to external stakeholders.

### **Course Outcome (CO) Bloom's Level**

Course outcomes (COs) are statements that describe the knowledge, skills, and abilities that students should possess and can demonstrate upon completion of a course. COs are typically written at the course level and are used to guide the development of the course and its assessments.

Bloom's Taxonomy is a framework for categorizing educational goals and objectives into different levels of complexity and specificity. The levels of Bloom's Taxonomy, from lowest to highest, are: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating.

The Bloom's level of a CO refers to the level of cognitive complexity that the CO is intended to address. For example, a CO that requires students to "recall the steps of the scientific method" would be at the Remembering level of Bloom's

Taxonomy, while a CO that requires students to “design an experiment to test a hypothesis” would be at the Creating level.

In summary: - Course outcomes (COs) describe the knowledge, skills, and abilities that students should possess upon completion of a course. - Bloom’s Taxonomy is a framework for categorizing educational goals and objectives. - The Bloom’s level of a CO refers to the level of cognitive complexity that the CO is intended to address.

**At the end of this course, the students should be able to:**

- Understand the fundamental concepts and principles of the subject.
- Apply the knowledge and skills acquired in the course to solve problems and make informed decisions.
- Communicate effectively using the language and terminology of the subject.
- Work collaboratively with others to achieve common goals.
- Demonstrate critical thinking and analytical skills.
- Conduct research and present findings in a clear and concise manner.
- Demonstrate ethical and responsible behavior in the application of the subject.
- Continue to learn and adapt to new developments in the field.

**CO 1 Conduct experiments illustrating the application of KVL/KCL and network theorems to DC electrical circuits. K3**

- KVL (Kirchhoff’s Voltage Law) states that the sum of the voltage drops around a closed loop in a circuit is equal to the sum of the voltage rises in that loop.
- KCL (Kirchhoff’s Current Law) states that the sum of the currents entering a node in a circuit is equal to the sum of the currents leaving that node.
- Network theorems, such as Thevenin’s theorem and Norton’s theorem, can be used to simplify complex circuits and make it easier to analyze them.
- To conduct experiments illustrating the application of KVL/KCL and network theorems to DC electrical circuits, one can set up a simple circuit with known resistances and voltage sources.
- Using a multimeter, one can measure the voltage drops and currents in the circuit and verify that they obey KVL and KCL.
- One can also use network theorems to simplify the circuit and predict the behavior of the circuit, and then verify the predictions through experimentation.
- These experiments can help students understand the fundamental principles of circuit analysis and the practical application of network theorems.

**CO 2 Demonstrate the behavior of AC circuits connected to single phase AC supply and measure power in single phase as well as three**



#### **phase electrical circuits. K4**

- An AC circuit is an electrical circuit that is powered by an alternating current (AC) source.
- A single-phase AC supply is an electric power supply that uses a single sinusoidal voltage waveform.
- In a single-phase AC circuit, the current and voltage vary sinusoidally with time.
- The behavior of an AC circuit connected to a single-phase AC supply can be analyzed using circuit analysis techniques such as Ohm's Law, Kirchhoff's Laws, and Thevenin's Theorem.
- Power in a single-phase AC circuit can be measured using a wattmeter, which measures the product of the voltage and current.
- A three-phase electrical circuit is a type of polyphase system that uses three conductors to carry three alternating currents of the same frequency but with a phase difference of 120 degrees.
- Power in a three-phase electrical circuit can be measured using a three-phase wattmeter, which measures the sum of the power in each of the three phases.
- The behavior of AC circuits and the measurement of power in single-phase and three-phase electrical circuits are important concepts in electrical engineering and are commonly studied in courses on circuit analysis and power systems.

#### **CO 3 Perform experiment illustrating BH curve of magnetic materials. K3**

The BH curve, also known as the hysteresis loop, is a graphical representation of the relationship between the magnetic field strength (H) and the magnetic flux density (B) of a magnetic material. The curve is used to characterize the magnetic properties of a material and can be used to predict its behavior in the presence of an external magnetic field.

To perform an experiment illustrating the BH curve of a magnetic material, the following steps can be followed:

1. Prepare a sample of the magnetic material to be tested. This can be in the form of a ring or a toroid.
2. Set up the test equipment, which typically includes a magnetizing coil, a flux meter, and a power supply.
3. Place the sample inside the magnetizing coil and connect the flux meter to measure the magnetic flux density (B) of the sample.
4. Gradually increase the current in the magnetizing coil to increase the magnetic field strength (H) applied to the sample.
5. Record the values of B and H as the current is increased.
6. Gradually decrease the current in the magnetizing coil to reduce the magnetic field strength (H) applied to the sample.

7. Record the values of B and H as the current is decreased.
8. Plot the recorded values of B and H on a graph to obtain the BH curve of the magnetic material.

The shape of the BH curve can provide valuable information about the magnetic properties of the material, such as its coercivity, remanence, and saturation magnetization. These properties can be used to predict the behavior of the material in the presence of an external magnetic field and can be useful in the design of magnetic devices and systems.

#### **CO 4 Calculate efficiency of a single phase transformer and DC machine. K4**

Efficiency is defined as the ratio of output power to input power. It is a measure of how effectively a device converts the power it receives into useful work.

To calculate the efficiency of a single phase transformer, the following steps can be taken:

1. Measure the input voltage and current to the transformer.
2. Measure the output voltage and current from the transformer.
3. Calculate the input power by multiplying the input voltage by the input current.
4. Calculate the output power by multiplying the output voltage by the output current.
5. Divide the output power by the input power to obtain the efficiency of the transformer.

Similarly, to calculate the efficiency of a DC machine, the following steps can be taken:

1. Measure the input voltage and current to the DC machine.
2. Measure the mechanical output power of the DC machine.
3. Calculate the input power by multiplying the input voltage by the input current.
4. Divide the mechanical output power by the input power to obtain the efficiency of the DC machine.

It is important to note that the efficiency of a transformer or a DC machine is not constant and can vary depending on the operating conditions. Factors such as load, temperature, and frequency can affect the efficiency of these devices. Therefore, it is important to measure the efficiency under the specific operating conditions for which the device is intended to be used.

#### **CO 5 Perform experiments on speed measurement and reversal of direction of three phase induction motor and Identify the type of DC and AC machines based on their construction. K4**

1. **Speed measurement of three phase induction motor:** The speed of a three phase induction motor can be measured using a tachometer, stroboscope, or by counting the number of revolutions per minute (RPM) using a stopwatch. The synchronous speed of the motor can be calculated using the formula  $N_s = 120f/P$ , where  $N_s$  is the synchronous speed,  $f$  is the frequency of the supply, and  $P$  is the number of poles.
2. **Reversal of direction of three phase induction motor:** The direction of rotation of a three phase induction motor can be reversed by interchanging any two of the three supply lines. This changes the direction of the rotating magnetic field, causing the rotor to rotate in the opposite direction.
3. **Identification of DC and AC machines based on their construction:** DC machines can be identified by the presence of a commutator, which is used to convert the alternating current (AC) produced in the armature windings into direct current (DC). AC machines, on the other hand, do not have a commutator and rely on the alternating nature of the supply to produce a rotating magnetic field. The construction of the rotor can also be used to differentiate between different types of AC machines, such as squirrel cage and wound rotor induction motors.

**K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create**

These are the six levels of Bloom's Taxonomy, a framework for categorizing educational goals and objectives. The taxonomy is hierarchical, meaning that each level builds upon the previous one.

1. **K1 – Remember:** This level involves recalling information from memory. Examples of tasks at this level include defining terms, listing facts, and reciting information.
2. **K2 – Understand:** This level involves comprehending the meaning of information. Examples of tasks at this level include explaining concepts, summarizing information, and interpreting data.
3. **K3 – Apply:** This level involves using information in a new situation. Examples of tasks at this level include solving problems, applying theories, and demonstrating techniques.
4. **K4 – Analyze:** This level involves breaking down information into parts and examining their relationships. Examples of tasks at this level include comparing and contrasting, categorizing, and identifying cause and effect.
5. **K5 – Evaluate:** This level involves making judgments based on criteria and standards. Examples of tasks at this level include evaluating the validity of arguments, assessing the quality of work, and making decisions.
6. **K6 – Create:** This level involves combining elements to form a new whole. Examples of tasks at this level include designing, constructing, and planning.

These levels can be used to design educational activities and assessments that promote higher-order thinking skills. By moving beyond simple recall and comprehension, students can develop deeper understanding and the ability to apply, analyze, evaluate, and create new knowledge.